

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 5: The Discrete-Time Fourier Transform (DTFT)

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1 The Discrete-Time Fourier Transform (DTFT)

Def.

The DTFT $X(e^{j\omega})$ of a sequence $x[n]$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}.^1 \quad (\text{Mitra 3.10})$$

$\Rightarrow$ a complex-valued function of a real-valued variable ω .

$$X(e^{j\omega}) = X_{\text{re}}(e^{j\omega}) + jX_{\text{im}}(e^{j\omega}).$$

Both $X_{\text{re}}(e^{j\omega})$ and $X_{\text{im}}(e^{j\omega})$ are real-valued functions of ω .

$$X(e^{j\omega}) = \underbrace{|X(e^{j\omega})|}_{\text{magnitude}} e^{j\theta(\omega)}, \quad \theta(\omega) = \angle X(e^{j\omega}) \text{ (phase)}.$$

¹Mitra: Equation (3.10) usually is referred to as the **analysis equation**, because it analyzes how much of each complex exponential signal is present in the original signal.

Remark.

Phase is not unique:

$$e^{j\theta(\omega)} = e^{j(\theta(\omega) + 2\pi k)}, \quad k \in \mathbb{Z}.$$

Sometimes we force the **principal value**, $-\pi \leq \theta(\omega) < +\pi$.

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DTFT Examples.

- $x[n] = \delta[n]$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\omega n} = \delta[0] e^{-j\omega \cdot 0} = 1.$$

- $x[n] = \alpha^n u[n]$, $|\alpha| < 1$:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} \alpha^n u[n] e^{-j\omega n} = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^n \\ &= \frac{1}{1 - \alpha e^{-j\omega}} \quad \text{if } |\alpha e^{-j\omega}| < 1 \quad \left(|\alpha e^{-j\omega}| = |\alpha| \underbrace{|e^{-j\omega}|}_1 \right). \end{aligned}$$

Example (Rectangular Window Sequence).

$$\begin{aligned} x[n] &= \begin{cases} 1, & 0 \leq n \leq N-1 \\ 0, & \text{else} \end{cases} \\ &= u[n] - u[n-N]. \end{aligned}$$

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$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{N-1} e^{-j\omega n} \\ &= \frac{1 - (e^{-j\omega})^N}{1 - e^{-j\omega}} \\ &= \frac{e^{-j\omega \frac{N}{2}} (e^{+j\omega \frac{N}{2}} - e^{-j\omega \frac{N}{2}})}{e^{-j\omega \frac{1}{2}} (e^{+j\omega \frac{1}{2}} - e^{-j\omega \frac{1}{2}})} \\ &= \frac{\sin(\omega \frac{N}{2})}{\sin(\omega \frac{1}{2})} e^{-j\omega \frac{N-1}{2}}. \end{aligned}$$

2 Dirichlet Kernel

In signal processing, the Dirichlet kernel is often called the periodic sinc function:

$$\frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})}.$$

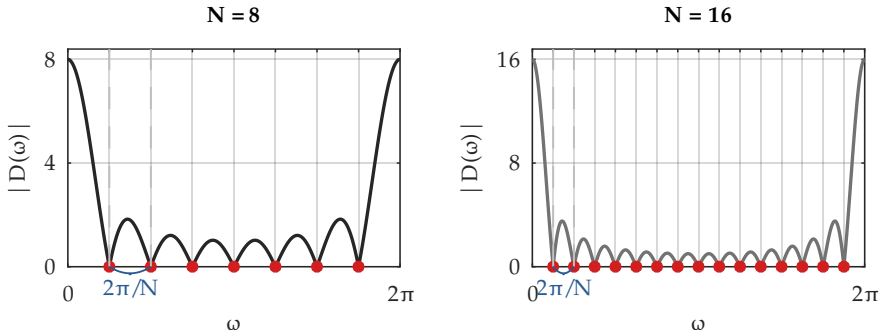
For small ω , $\sin(\frac{\omega}{2}) \approx \frac{\omega}{2}$, so

$$\frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})} \approx \frac{\sin(\omega \frac{N}{2})}{\frac{\omega}{2}} = N \operatorname{sinc}\left(\omega \frac{N}{2}\right), \quad \operatorname{sinc}(x) \triangleq \frac{\sin x}{x}.$$

Zeros occur when $\omega \frac{N}{2} = k\pi$, $k \neq 0$, i.e., $\omega = \frac{2\pi k}{N}$.



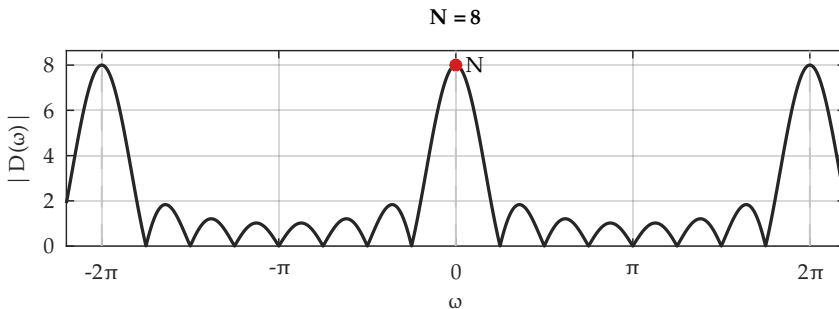
P. G. Lejeune Dirichlet



“more zeros” for larger N , faster oscillations.

4 But: also periodic (“periodic sinc”):

$$\left| \frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})} \right| \stackrel{?}{=} \left| \frac{\sin((\omega + 2\pi) \frac{N}{2})}{\sin(\frac{\omega + 2\pi}{2})} \right| = \left| \frac{\sin(\omega \frac{N}{2} + \pi N)}{\sin(\frac{\omega}{2} + \pi)} \right|.$$



$$|D(\omega)| = |D(\omega + 2\pi)|$$

Dealing with $0/0$ when $\omega = 0$:

$$X(e^{j\omega}) \Big|_{\omega=0} = \sum_{n=0}^{N-1} e^{-j\omega n} \Big|_{\omega=0} = N.$$

Limits: $\lim_{\omega \rightarrow 0} \frac{f(\omega)}{g(\omega)}$. For small ω , $\frac{\omega \frac{N}{2}}{\frac{\omega}{2}} = N$.

5 3 DTFT Properties

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3.1 Periodicity

Unlike the continuous-time Fourier transform, the DTFT is a periodic function of ω with period 2π .³

Property.

The DTFT is 2π -periodic: $X(e^{j(\omega+2\pi k)}) = X(e^{j\omega})$ for all $k \in \mathbb{Z}$.

Proof.

$$\begin{aligned} X(e^{j(\omega_0+2\pi k)}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega_0+2\pi k)n} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega_0 n} \underbrace{e^{-j2\pi kn}}_{e^{-j2\pi kn}=1} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega_0 n} = X(e^{j\omega_0}), \quad \text{for all values of } k. \quad \blacksquare \end{aligned}$$

²cf. Mitra Sec. 3.2.2, Basic Properties.

³cf. Mitra Sec. 3.2.1.

Remark*(addendum to orig lecture note).*

Why is $e^{-j2\pi kn} = 1$? Both k and n are integers, so $kn \in \mathbb{Z}$ and the angle $-2\pi kn$ is an integer multiple of 2π . Each full turn on the unit circle returns to 1:

$$e^{-j2\pi kn} = \cos(2\pi kn) - j \sin(2\pi kn) = 1.$$

3.2 Inverse transform**Inverse DTFT.**

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{+j\omega n} d\omega. \quad (Mitra 3.14)$$

Equations (3.10) and (3.14) form a DTFT pair. Write

$$\mathcal{F}\{x[n]\} = X(e^{j\omega}), \quad \mathcal{F}^{-1}\{X(e^{j\omega})\} = x[n],$$

and denote the DTFT pair by

$$x[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega}). \quad (Mitra 3.15)$$

⁴Mitra:

- It should be noted that even though the integration in Eq. (3.14) can be carried out over any interval of duration 2π , it is a common practice to choose the interval $[-\pi, \pi]$.
- The inverse discrete-time Fourier transform given by Eq. (3.14) can be interpreted as a linear combination of infinitesimally small complex exponential signals of the form $\frac{1}{2\pi} e^{j\omega n} d\omega$, weighted by the complex constant $X(e^{j\omega})$ over the normalized angular frequency range from $-\pi$ to π .
- Eq. (3.14) is referred to as the **synthesis equation**, because it synthesizes an arbitrary signal from its complex exponential components.

Proof.

$$\begin{aligned}
 x[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_{\ell=-\infty}^{\infty} x[\ell] e^{-j\omega\ell} \right) e^{j\omega n} d\omega \\
 &= \sum_{\ell=-\infty}^{\infty} x[\ell] \int_{-\pi}^{\pi} \frac{1}{2\pi} e^{j\omega(n-\ell)} d\omega \\
 &= \sum_{\ell=-\infty}^{\infty} x[\ell] \frac{\sin(\pi(n-\ell))}{\pi(n-\ell)} = \sum_{\ell=-\infty}^{\infty} x[\ell] \delta[n-\ell] \\
 &= x[n].
 \end{aligned}$$

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3.3 Convergence

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Remark.

Exchanging the order of integration and summation (above) assumed convergence.

Convergence.

An infinite series of the form $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$ (3.10) may or may not converge. Let

$$X_K(e^{j\omega}) = \sum_{n=-K}^K x[n] e^{-j\omega n}. \quad (\text{Mitra 3.24})$$

denote the partial sum of the weighted complex exponentials in Eq. (3.10), and define

$$X(e^{j\omega}) = \lim_{K \rightarrow \infty} X_K(e^{j\omega})$$

when this limit exists (in the sense made precise below).

⁵cf. Mitra Sec. 3.2.4.

Def.

We say that $X_K(e^{j\omega})$ converges **uniformly** to $X(e^{j\omega})$ —stronger than point-wise convergence—if for every $\varepsilon > 0$, there exists an integer K such that for all integers $L \geq K$, the partial sum $X_L(e^{j\omega})$ of Eq. (3.24) satisfies

$$|X(e^{j\omega}) - X_L(e^{j\omega})| < \varepsilon \quad \text{for all } \omega.$$

Property.

If $x[n]$ is an **absolutely summable** sequence—that is, if Eq. (3.25) holds with

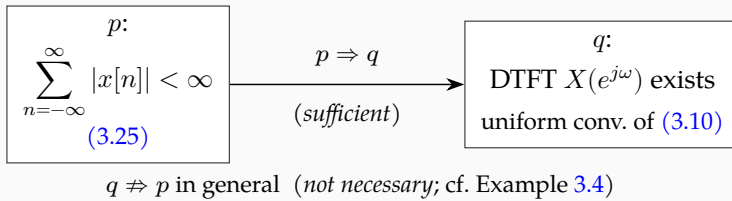
$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty. \quad (\text{Mittra 3.25})$$

then, by the triangle inequality and $|e^{-j\omega n}| = 1$,

$$|X(e^{j\omega})| = \left| \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |x[n]| |e^{-j\omega n}| = \sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

for all ω , so the DTFT $X(e^{j\omega})$ of $x[n]$ exists for every ω .

Eq. (3.25) is thus a **sufficient condition** for existence of the DTFT $X(e^{j\omega})$:



Moreover, since $|x[n]e^{-j\omega n}| = |x[n]|$ is independent of ω , the Weierstrass M -test shows that the series of Eq. (3.10) converges **uniformly** for all ω .

Def.

We say that $X_K(e^{j\omega})$ converges in the **mean-square** to $X(e^{j\omega})$ if

$$\lim_{K \rightarrow \infty} \int_{-\pi}^{\pi} |X(e^{j\omega}) - X_K(e^{j\omega})|^2 d\omega = 0.$$

Property.

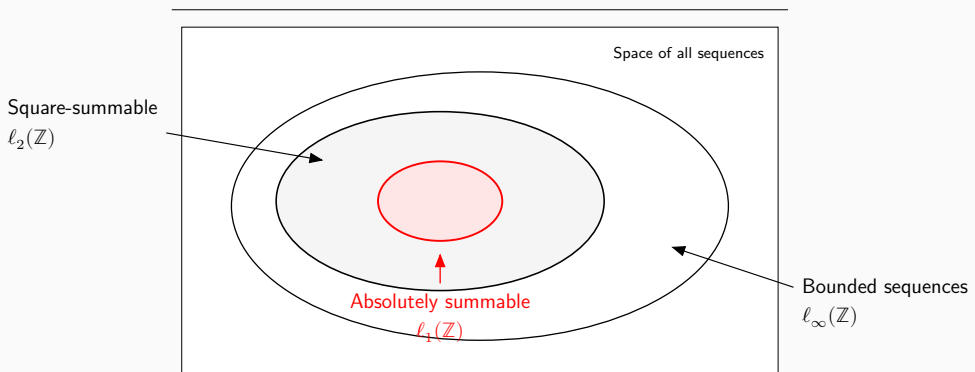
The DTFT will converge in the mean-square if the sequence $x[n]$ is **square-summable**:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty.$$

Property (Nesting of the $\ell_p(\mathbb{Z})$ spaces).

- Every absolutely summable sequence is also square summable.
- Every square-summable sequence is bounded.

$$\ell_1(\mathbb{Z}) \subset \ell_2(\mathbb{Z}) \subset \ell_\infty(\mathbb{Z}).$$



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3.4 An Example, and Generalized DTFTs

Example.

The sequence

$$x[n] = \begin{cases} 1/n, & n \geq 1 \\ 0, & n \leq 0 \end{cases}$$

is **not** absolutely summable, but **is** square summable (famous result by Euler):

$$\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^2 = \frac{\pi^2}{6}.$$

So $x[n] \in \ell_2(\mathbb{Z}) \setminus \ell_1(\mathbb{Z})$: absolute summability is **not necessary** for a sequence to admit a useful DTFT analysis in the mean-square sense (cf. the diagram in §3.3).

We can sometimes define DTFTs in cases where the DTFT does not actually converge (but still exists in a generalized sense).

Example.

$$x[n] = e^{j\omega_0 n}:$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} e^{j(\omega_0 - \omega)n} \quad \text{does not converge.}$$

But $x[n]$ still has a useful (generalized) DTFT:

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 + 2\pi k).$$

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$$X(e^{j\omega}) = 2\pi \underbrace{\delta(\omega - \omega_0)}_{\text{Dirac delta function}}, \quad -\pi \leq \omega \leq \pi, \quad -\pi \leq \omega_0 \leq \pi$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} 2\pi \delta(\omega - \omega_0) e^{j\omega n} d\omega = e^{j\omega_0 n}.$$

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 4.8
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 2.7